

Functional genetic report · longevity

Advanced Cardiovascular

Sérgio F · Report date: 02-10-2026

Lp(a) and KIV-2, coagulation and thrombosis, 9p21 locus, coronary artery disease, atrial fibrillation, QT interval, aortic aneurysm, heart failure and stroke. Only genes (chip + local imputation + polygenic scores).

Contents

• Summary	3
• How to read this report	5
• Polygenic scores	6
• Lipoprotein(a): LPA and KIV-2 proxies	7
• Coagulation and venous thrombosis	8
• 9p21 locus	10
• Other loci of coronary disease	11
• Atrial fibrillation	13
• QT interval and arrhythmias	14
• Aortic aneurysm	15
• Cardiac and myocardial insufficiency	16
• Stroke	17
• Mechanisms of ageing	18
• Practical suggestions	19
• Limitations	20
• Methods and sources	21

Summary

83

SNPs analyzed

74

in the chip

5

imputed (DR2 $\geq$ 0.8)

4

SNP not found in the data

12

Polygenic scores

This report combines, in a single site, the cardiovascular genetics of Sérgio F that goes beyond cholesterol: lipoprotein(a), coagulation, 9p21 locus, non-lipidic coronary loci, heart rate (atrial fibrillation and QT interval), aorta, cardiac muscle and stroke. The conclusions come only from genes.

In summary: vascular genetics is mixed. There are two clear and LDL-independent weaknesses (risk homozygous 9p21 and a moderate polygenic predisposition for venous thrombosis, without Factor V Leiden or prothrombin), and several clear forces: no high Lp(a) alleles, low global coronary score (metaGRS at the 5th percentile), very low genetic risk of atrial fibrillation (the 2nd percentile), probable blood group O, **APOE ϵ 3/ ϵ 3** and low scores for stroke and heart failure.

Favourable

- **Lp(a):** without any of the high Lp(a) alleles (LPA rs10455872 AA, rs3798220 TT, rs140570886 TT, rs118039278 GG, rs73596816 GG, rs186696265 CC).
- **Coronary disease:** metaGRS (1.7 M variants) at the 5th percentile; GPS_Mult in 30. PHACTR1 rs9349379 AA (protector), IL6R Asp358Ala heterozygous (less IL-6 signal), LDLR rs6511720 GT, APOE ϵ 3/ ϵ 3.
- **Coagulation:** Factor V Leiden and prothrombin G20210A negative; ABO rs505922 TT (probable group O: less factor VIII/von Willebrand).
- **Atrial fibrillation:** Score at the 2nd percentile; without risk alleles in PITX2 (3 SNPs), ZFX3 and PRRX1.
- **Stroke and heart failure:** scores at the 19 and 22 percentiles; BAG3 without risk allele; TTR V122I and V30M absent.
- **Aorta and QT:** Aneurysm score at the 26th percentile (DAB2IP without risk); QT at the 36th percentile (NOS1AP without alleles that extend).

Unfavourable

- **9p21 homozygous risk** (rs1333049 CC, rs10757278 GG, rs4977574 GG, rs10757274 GG, rs2383206 GG): the strongest common genetic factor of coronary and aneurysmal disease, regardless of LDL ($\sim$ 1.5–1.7 $\times$ vs. no risk alleles).
- **Venous thrombosis** 297 score variants at the 86th percentile; F11 rs2289252 TT (higher XI factor), with F11 rs2036914 TC and FGG rs2066865 GA to add.
- **Other coronary loci:** APOA5 rs964184 GC (infrequent risk allele; triglycerides) and TCF21 rs12190287 CC; 27 SNPs score at the 80th percentile (pulled by 9p21). CXCL12 CC and GUCY1A1 GG are at risk, but they are the most common (typical) genotypes.
- **Rare protective variants absent** (without PCSK9 R46L, APOC3 R19X, ANGPTL4 E40K or LPL S447X): it is not a risk, but there is no extra genetic shield.
- **QT and rhythm:** PLN rs11970286 TT (prolongs slightly the QT) and TTN rs2288327 AG (atrial fibrillation, small effect); KCNE1 D85N not found in the data.

How these pieces combine: The global coronary score (million variants) is low despite 9p21, because the rest of the genome compensates. 9p21 acts mainly on the arterial wall (smooth muscle proliferation, senescence via CDKN2A/B), not on cholesterol. Therefore, the strategy that yields the most is the same as always: LDL/ApoB as low as possible (the lower, the better) and much fiber, vegetables and fruit, which in the literature attenuate the effect of 9p21.

Polygenic scores update (03-10-2026). The scores were recalculated with more published recent scores (the same method and the same reference population). This box prevails over the previous values cited in the text.

- Coronary disease: new scores p54, **p18** and p5.5; with the previous ones, the consensus is p18. It remains low, less extreme than p5.
- Venous thrombosis: 3 new scores give p24 and p71 (genomic) and p80 (10 SNPs); with the previous ones (p86, p84 and p54), the consensus is **p76**. It goes from high to above average; the practical rules remain.
- Atrial fibrillation: new p9, p11 and p3 scores; confirms low.

How to read this report

Genetics only. All conclusions come only from Sérgio F genes (without blood tests or exams). They are predispositions, non-diagnosis. Each theme is divided into favourable and unfavourable; what does not have clear judgment appears as typical.

Strand: All genotypes are in the strand + of dbSNP (GRCh38). In the SNPs of this report, 75 directly agree with the dbSNP alleles, 0 needed conversion to the complementary strand and 1 are palindromic (A/T or C/G; orientation confirmed by the GRCh38 position). The genotypes are already in the strand + of the reference. Therefore, some genotypes appear with different letters from those used in the literature when using the strand of the gene (e.g. MTHFR C677T = rs1801133 A in the strand +). Disagreements (excluded): rs8176719.

Origin and trust: chip = directly genotyped (high confidence); imputation DR2=x = locally inferred from a public panel of complete genomes (30x), accepted only with $DR2 \geq 0.8$ (good trust, just below the chip); SNP not found in the data : neither chip nor imputed with sufficient quality (indels, rare variants, deletions).

Effect and judgment allele: The comparison between the number of Sérgio F copies and the mean of the reference population ($2 \times$ allele frequency; column «Freq.»). **Unfavourable** = at least 0.6 copies of risk allele above average (or any copy of a rare variant of strong effect); **Favourable** = at least 0.6 below average, protective allele, or absence of a strong effect variant; **Typical / no judgement** = close to the average (e.g. homozygous for a risk allele that is the most common) or context variant. Each common SNP has a small individual effect; what weighs more are the variants of large effect (marked as strong evidence) and the polygenic scores.

Polygenic scores (PGS): add thousands of variants with coefficients published in PGS Catalog. The percentile compares Sérgio F with the reference population (n=633), calculated with exactly the same variants. Percentage ≥ 80 in a risk score = unfavourable; ≤ 20 = favourable.

Polygenic scores

Coronary disease and Lp(a) scores were reused from genomic imputation (even method). The remaining ones were now calculated. Percentage ≥ 80 = unfavourable; ≤ 20 = favourable.

Trait	Score	Variables used (coverage)	Percentage (reference population)	Judgement
Coronary disease (metaGRS, 1.7 M variants)	PGS000018	1 476 469 of 1 745 179 (85%)	p5	Favourable
Coronary disease (GPS_Mult, 1.3 M variants)	PGS003725	1 116 308 of 1 296 172 (86%)	p30	Typical / no judgement
Coronary disease (27 SNPs, Mega 2015)	PGS000010	26 of 27 (96%)	p80	Unfavourable
Lp(a) (43 variants)	PGS000667	28 of 43 (65%)	p61	Typical / no judgement
Lp(a) (11 446 variants)	PGS000752	10 808 of 11 446 (94%)	p75	Typical / no judgement
Venous thrombosis (297 variants, Klarin 2019)	PGS000043	265 of 297 (89%)	p86	Unfavourable
Venous thrombosis (36 variants)	PGS002235	32 of 36 (89%)	p63	Typical / no judgement
Atrial fibrillation (166 variants)	PGS004920	163 of 166 (98%)	p2	Favourable
QT interval (68 variants)	PGS000768	65 of 68 (96%)	p36	Typical / no judgement
Abdominal aneurysm (29 variants)	PGS000753	27 of 29 (93%)	p26	Typical / no judgement
Heart failure (39 variants)	PGS003969	38 of 39 (97%)	p22	Typical / no judgement
Stroke (90 variants)	PGS000038	84 of 90 (93%)	p19	Favourable

Lipoprotein(a): LPA and KIV-2 proxies

Lp(a) is almost all genetic (70–90%) and is a causal factor of coronary disease and aortic stenosis. The main determinant is the number of kringle repetitions IV type 2 (KIV-2) of the LPA gene, which the chips do not measure; the SNPs below are markers (proxies) of small and Lp(a) high isoforms.

Polygenic scores of this theme: Lp(a) (43 variants) (PGS000667): the 61st percentile **Typical / no judgement** ; Lp(a) (11 446 variants) (PGS000752): the 75th percentile **Typical / no judgement**

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
LPA	rs10455872	AA	chip	G	0	0.07	Main SNP of high Lp(a) (small isoforms of apo(a)); ~1.5× coronary risk per copy. (strong evidence)	Favourable
LPA	rs3798220	TT	chip	C	0	0.01	LPA I4399M: C allele associated with very high Lp(a) and coronary/aortic risk. (strong evidence)	Favourable
LPA	rs41272114	CC	chip	T	0	0.04	Splicing variant (zero allele): low to Lp(a). (moderate evidence)	Typical / no judgement
LPA	rs140570886	TT	chip	C	0	0.01	LPA variant associated with Lp(a) (proxy of isoforms/KIV-2). (moderate evidence)	Typical / no judgement
LPA	rs6415084	CC	chip	T	0	0.48	LPA variant linked to the number of repetitions KIV-2 (proxy). (weak evidence)	Favourable
LPA	rs118039278	GG	imputation DR2=1.00	A	0	0.07	LPA variant associated with Lp(a) (in LD with rs10455872). (strong evidence)	Favourable
LPA	rs186696265	CC	chip	T	0	0.01	Variant near LPAL2 associated with Lp(a). (moderate evidence)	Typical / no judgement
LPA	rs73596816	GG	chip	A	0	0.03	LPA variant associated with Lp(a). (moderate evidence)	Typical / no judgement

Favourable

- LPA rs10455872 AA
- LPA rs3798220 TT
- LPA rs6415084 CC
- LPA rs118039278 GG

Unfavourable

- None.

Interpretation: None of the 7 markers of Lp(a) high is present, which makes Lp(a) very high unlikely. The polygenic scores of Lp(a) are in the percentiles 61 and 75 (typical). As KIV-2 is not visible in the chip, genetics does not completely exclude a high Lp(a); the genetic plan already deals with the unique measurement of Lp(a).

Coagulation and venous thrombosis

Classical hereditary thrombophilias (Factor V Leiden, prothrombin G20210A) and common variants that add risk of venous thrombosis (XI factor, fibrinogen, ABO group). Protein C, protein S and antithrombin deficits are rare and are not covered by the chip.

Polygenic scores of this theme: Venous thrombosis (297 variants, Klarin 2019) (PGS000043): the 86th percentile **Unfavourable**; Venous thrombosis (36 variants) (PGS002235): the 63rd percentile

Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
F5 Factor V Leiden (R506Q)	rs6025	CC	chip	T	0	0.01	Factor V Leiden: resistance to activated protein C; ~3-7× risk of venous thrombosis by copy. (strong evidence)	Favourable
F2 Prothrombin G20210A	rs1799963	GG	chip	A	0	0.01	Prothrombin G20210A: more prothrombin; ~2-3× risk of venous thrombosis. (strong evidence)	Favourable
F11	rs2289252	TT	chip	T	2	0.39	Higher XI factor; risk of venous thrombosis. (moderate evidence)	Unfavourable
F11	rs2036914	TC	chip	C	1	0.54	XI factor; risk of venous thrombosis. (moderate evidence)	Typical / no judgement
FGG	rs2066865	GA	chip	A	1	0.22	Gamma fibrinogen (γ): risk of venous thrombosis. (moderate evidence)	Typical / no judgement
FGA Thr312Ala	rs6050	TC	chip	C	1	0.23	Fibrinogen alpha: more rigid clot; thrombosis/embolism. (weak evidence)	Typical / no judgement
FGB -455G>A	rs1800790	GA	chip	A	1	0.21	Beta fibrinogen promoter: higher plasma fibrinogen. (weak evidence)	Typical / no judgement
ABO non-O group (tag)	rs505922	TT	chip	C	0	0.37	Tag of the non-O blood group: plus factor VIII/von Willebrand, plus venous thrombosis and infarction. (moderate evidence)	Favourable
ABO non-O group (tag)	rs8176719	—	SNP not found in the data	T	—	—	Deletion of the O allele (indel; chips rarely read it). Indel of the O allele: the chip returns a code that does not correspond to the dbSNP alleles; the rs505922 tag is used. (moderate evidence)	SNP not found in the data
F13A1 Val34Leu	rs5985	CA	chip	A	1	0.24	XIII Factor: Leu34 forms crosslinks faster. (weak evidence)	Typical / no judgement
SERPINE1 PAI-1 4G/5G	rs1799889	—	SNP not found in the data	—	—	—	Indel of the PAI-1 promoter (4G = more PAI-1, less fibrinolysis). Indel: rarely read on the chip. Indel 4G/5G: the chip returns a code that does not allow to read the genotype reliably. (weak evidence)	SNP not found in the data
MTHFR C677T	rs1801133	GA	chip	A	1	0.37	Background: T allele (A in strand +) reduces the activity of MTHFR and rises homocysteine; small thrombotic effect. (weak evidence)	Typical / no judgement
MTHFR A1298C	rs1801131	TG	chip	G	1	0.31	Background: C allele (G at strand +) slightly reduces MTHFR activity. (weak evidence)	Typical / no judgement
SH2B3 R262W	rs3184504	CC	chip	T	0	0.46	SH2B3/LNK: more platelets and inflammation; coronary risk and thrombosis. (moderate evidence)	Favourable

PROC/ PROS1/ SERPINC1	rs121918145	SNP not found in the data	—	—	—	Protein C/S/antithrombin deficits: rare variants; the chip does not cover them well. (strong evidence)	SNP not found in the data
-----------------------------	-------------	---------------------------------	---	---	---	--	------------------------------

Favourable

- Factor V Leiden (rs6025 CC) and prothrombin G20210A (rs1799963 GG): negative.
- ABO rs505922 TT: probable group O (less factor VIII and von Willebrand; less venous thrombosis and infarction).
- SH2B3 R262W without the risk allele (rs3184504 CC).

Unfavourable

- F11 rs2289252 **TT**: higher factor XI (each allele ~1.2–1.4× risk of venous thrombosis).
- F11 rs2036914 TC and FGG rs2066865 GA: 1 copy each (close to the average, but add).
- Venous thrombosis score (297 variants) at the 86th percentile.

Interpretation: No major hereditary thrombophilia, but with a moderate polygenic predisposition to venous thrombosis (F11 + FGG + score at the 86th percentile). In practice, it is a risk that only manifests itself with triggers (prolonged immobilization, surgery, dehydration, long journeys). MTHFR C677T GA and A1298C TG appear as a context: they have a small thrombotic effect and are already treated in the subject of methylation. The PAI-1 4G/5G indels and the ABO O allele are not read reliably by the chip.

9p21 locus

The 9p21.3 locus (ANRIL/CDKN2B-AS1, with CDKN2A/CDKN2B, that encode p16INK4a and p15INK4b) is the strongest common sign of coronary disease. It acts in the arterial wall (proliferation and senescence of vascular smooth muscle, inflammation), not in cholesterol. It is also the main common locus of abdominal aortic aneurysm.

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
CDKN2B-AS1 (9p21)	rs1333049	CC	chip	C	2	0.48	Locus 9p21 (ANRIL/CDKN2A/B): ~1.25–1.3× coronary risk per copy, independent of lipids; also aortic aneurysm. (moderate evidence)	Unfavourable
CDKN2B-AS1 (9p21)	rs10757278	GG	chip	G	2	0.48	9p21: coronary and aneurysm risk. (moderate evidence)	Unfavourable
CDKN2B-AS1 (9p21)	rs4977574	GG	chip	G	2	0.50	9p21: coronary risk (leading variable in several GWAS). (moderate evidence)	Unfavourable
CDKN2B-AS1 (9p21)	rs10757274	GG	chip	G	2	0.50	9p21: coronary risk. (moderate evidence)	Unfavourable
CDKN2B-AS1 (9p21)	rs2383206	GG	chip	G	2	0.51	9p21: coronary risk. (moderate evidence)	Unfavourable

Favourable

- None.

Unfavourable

- CDKN2B-AS1 (9p21) rs1333049 CC
- CDKN2B-AS1 (9p21) rs10757278 GG
- CDKN2B-AS1 (9p21) rs4977574 GG
- CDKN2B-AS1 (9p21) rs10757274 GG
- CDKN2B-AS1 (9p21) rs2383206 GG

Interpretation: Sérgio F is homozygous for the risk haplotype in 5 SNPs (all genotyped in the chip, concordant). For those who have 2 copies, the relative risk of coronary disease is ~1.5–1.7× that of those who do not have any. The effect is independent of LDL, but studies show that a diet rich in raw vegetables and fruit attenuates the risk of 9p21 (Do et al., 2011). It binds directly to cellular senescence (p16INK4a).

Other loci of coronary disease

Loci of coronary disease outside 9p21 (lipids and non-lipids), rare protective variants and APOE. The global coronary scores include these loci and millions of others.

Polygenic scores of this theme: Coronary disease (metaGRS, 1.7 M variants) (PGS000018): the 5th percentile

Favourable; Coronary disease (GPS_Mult, 1.3 M variants) (PGS003725): the 30th percentile

Typical / no judgement; Coronary disease (27 SNPs, Mega 2015) (PGS000010): the 80th percentile **Unfavourable**

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
PCSK9 R46L	rs11591147	GG	chip	T	0	0.02	Loss of PCSK9 function: Lower LDL for life, ~30-50% less coronary disease. (strong evidence)	Typical / no judgement
LDLR	rs6511720	GT	chip	T	1	0.11	Allele T lowers LDL and coronary and aneurysm risk. (moderate evidence)	Favourable
CELSR2/ SORT1	rs12740374	GT	chip	G	1	0.79	Locus 1p13 (SORT1): risk allele rises LDL and coronary risk. (moderate evidence)	Typical / no judgement
CELSR2/ SORT1	rs599839	GA	chip	A	1	0.78	Locus 1p13 (SORT1): LDL and coronary risk. (moderate evidence)	Typical / no judgement
PHACTR1	rs9349379	AA	chip	G	0	0.39	PHACTR1/EDN1: coronary risk and calcification. (moderate evidence)	Favourable
IL6R Asp358Ala	rs2228145	AC	chip	C	1	0.36	Ala358 (C): less IL-6 signaling (tocilizumab type effect); less coronary disease and aneurysm. (moderate evidence)	Favourable
ADAMTS7	rs3825807	GG	chip	A	0	0.56	ADAMTS7 (mild muscle migration): coronary risk. (moderate evidence)	Favourable
ZC3HC1	rs11556924	CT	chip	C	1	0.62	ZC3HC1: coronary risk. (weak evidence)	Typical / no judgement
MIA3	rs17465637	CC	chip	C	2	0.74	MIA3: coronary risk. (weak evidence)	Typical / no judgement
TCF21	rs12190287	CC	imputation DR2=0.93	C	2	0.62	TCF21 (fibroblasts / smooth muscle): coronary risk. (weak evidence)	Unfavourable
CXCL12	rs1746048	CC	chip	C	2	0.85	CXCL12: coronary risk. (weak evidence)	Typical / no judgement
COL4A1/ COL4A2	rs4773144	AG	chip	G	1	0.45	IV collagen of the vascular basal membrane: coronary risk. (weak evidence)	Typical / no judgement
APOA5/ ZPR1	rs964184	GC	chip	G	1	0.15	Locus APOA5: higher triglycerides and coronary risk. (moderate evidence)	Unfavourable
LPL S447X	rs328	CC	chip	G	0	0.14	LPL S447X (function gain): lower triglycerides, less coronary disease. (moderate evidence)	Typical / no judgement
ANGPTL4 E40K	rs116843064	GG	chip	A	0	0.03	ANGPTL4 E40K: lower triglycerides, less coronary disease (rare). (moderate evidence)	Typical / no judgement
APOC3 R19X	rs76353203	CC	chip	T	0	—	APOC3 R19X (null): very low triglycerides, ~40% less coronary disease. (strong evidence)	Typical / no judgement

GUCY1A1	rs7692387	GG	chip	G	2	0.80	Soluble guanylate cyclase (NO pathway): coronary risk. (weak evidence)	Typical / no judgement
APOE ε2/ε3/ε4 (rs429358)	rs429358	TT	imputation DR2=0.99	C	0	0.15	Allele C = ε4: LDL and higher coronary risk. (strong evidence)	Favourable
APOE ε2/ε3/ε4 (rs7412)	rs7412	CC	chip	T	0	0.07	Allele T = ε2: Lower LDL (with no effect ε3/ε3). (moderate evidence)	Typical / no judgement

Favourable

- metaGRS (1.7 M variants) at the 5th percentile; GPS_Mult in 30.
- PHACTR1 rs9349379 AA (double dose protective allele).
- IL6R Asp358Ala AC (less IL-6 signal; tocilizumab type effect).
- LDLR rs6511720 GT (1 allele that lowers LDL).
- APOE ε3/ε3 (without ε4); ADAMTS7 GG and SH2B3 CC without risk allele.

Unfavourable

- APOA5 rs964184 GC (infrequent risk allele; triglycerides).
- TCF21 rs12190287 CC (imputed).
- 27 SNPs score at the 80th percentile (pulled by 9p21).
- Typical (most common, risk genotype): CXCL12 CC, GUCY1A1 GG, MIA3 CC. No rare protective variants (PCSK9 R46L, APOC3 R19X, ANGPTL4 E40K, LPL S447X).

Interpretation: The global score (the most informative, with 1.7 million variants) is low: the rest of the genome largely compensates for 9p21. The scores of a few dozen SNPs overestimate the 9p21 and therefore give a higher percentile. Conclusion: Polygenic coronary risk below average, with a localized weakness (9p21) in the arterial wall.

Atrial fibrillation

Atrial fibrillation is the most common arrhythmia of aging and one of the main causes of stroke. The 4q25 locus (PITX2) is its strongest genetic signal.

Polygenic scores of this theme: Atrial fibrillation (166 variants) (PGS004920): the 2nd percentile Favourable

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
PITX2 (4q25)	rs2200733	CC	chip	T	0	0.15	Locus 4q25/PITX2, the largest genetic signal of atrial fibrillation (~1.6x per copy). (moderate evidence)	Typical / no judgement
PITX2 (4q25)	rs6843082	AA	chip	G	0	0.25	4q25/PITX2: atrial fibrillation. (moderate evidence)	Typical / no judgement
PITX2 (4q25)	rs6817105	TT	chip	C	0	0.15	4q25/PITX2: atrial fibrillation. (moderate evidence)	Typical / no judgement
ZFHX3	rs2106261	CC	chip	T	0	0.17	ZFHX3: atrial fibrillation. (moderate evidence)	Typical / no judgement
ZFHX3	rs7193343	CC	chip	T	0	0.17	ZFHX3: atrial fibrillation. (weak evidence)	Typical / no judgement
KCNN3	rs13376333	CT	chip	T	1	0.30	KCNN3 potassium channel: atrial fibrillation. (weak evidence)	Typical / no judgement
PRRX1	rs3903239	AA	chip	G	0	0.42	PRRX1: atrial fibrillation. (weak evidence)	Favourable
SYNE2	rs1152591	AG	chip	A	1	0.46	SYNE2 (nesprine; nuclear lamina): atrial fibrillation. (weak evidence)	Typical / no judgement
SYNPO2L	rs10824026	AA	chip	A	2	0.85	SYNPO2L: atrial fibrillation. (weak evidence)	Typical / no judgement
NEURL1	rs12415501	CC	chip	T	0	0.13	NEURL1: atrial fibrillation. (weak evidence)	Typical / no judgement
TTN	rs2288327	AG	chip	G	1	0.17	Titin (TTN): atrial fibrillation. (weak evidence)	Unfavourable

Favourable • PRRX1 rs3903239 AA	Unfavourable • TTN rs2288327 AG
--	--

Interpretation: Very low genetic risk: score at percentile 2 and no allele of risk in 3 SNPs of PITX2, in ZFHX3 and in PRRX1. TTN rs2288327 AG is a small exception (SYNPO2L AA is the most common genotype). Zero alcohol, which Sérgio F already fulfills, is the main modifiable factor.

QT interval and arrhythmias

The QT interval reflects ventricular repolarization. Long QT (genetic or drug-induced) predisposes to severe arrhythmias. The monogenic channelopathies (KCNQ1, KCNH2, SCN5A rare) are not covered by the chip.

Polygenic scores of this theme: QT interval (68 variants) (PGS000768): the 36th percentile

Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
NOS1AP	rs12143842	CC	chip	T	0	0.25	NOS1AP: the largest common sign of QT interval (prolonged ~3 ms per copy). (moderate evidence)	Typical / no judgement
NOS1AP	rs10494366	TT	chip	G	0	0.37	NOS1AP: interval QT. (weak evidence)	Favourable
KCNH2 K897T	rs1805123	TG	chip	T	1	0.75	HERG channel (KCNH2): QT interval. (weak evidence)	Typical / no judgement
KCNQ1	rs12296050	CT	chip	T	1	0.23	KCNQ1: interval QT. (weak evidence)	Typical / no judgement
KCNQ1	rs2074238	CC	chip	C	2	0.93	KCNQ1: rare allele shortens QT. (moderate evidence)	Typical / no judgement
KCNE1 D85N	rs1805128		SNP not found in the data	T	—	—	KCNE1 D85N: prolongs QT and increases the risk of long-term drug-induced QT (~1–2% of the reference population). (moderate evidence)	SNP not found in the data
SCN5A H558R	rs1805124	TT	chip	C	0	0.21	Cardiac sodium channel SCN5A: conduction (PR/QRS). (weak evidence)	Typical / no judgement
SCN5A S1103Y	rs7626962	GG	chip	T	0	0.00	SCN5A S1103Y: arrhythmia (rare in the reference population). (moderate evidence)	Typical / no judgement
PLN	rs11970286	TT	chip	T	2	0.45	Phosfolambane (PLN; reticulum calcium): QT interval. (weak evidence)	Unfavourable
ATP1B1	rs10919071	AA	chip	A	2	0.87	Na/K (ATP1B1) pump: QT interval. (weak evidence)	Typical / no judgement
LITAF	rs8049607	TC	chip	T	1	0.51	LITAF: interval QT. (weak evidence)	Typical / no judgement

Favourable

- NOS1AP rs10494366 TT

Unfavourable

- PLN rs11970286 TT

Interpretation: QT score at the 36th percentile (typical) and NOS1AP without alleles that extend the QT. PLN TT slightly prolongs QT (some ms); ATP1B1 AA is also the most common genotype. **KCNE1 D85N**, the common variant that most increases the risk of long QT with drugs, was not found in the data: it is unknown. SCN5A S1103Y absent; H558R TT (common genome).

Aortic aneurysm

Abdominal aneurysm: degeneration of the aorta wall (elastin, collagen, inflammation). Shared loci with coronary disease (9p21, LDLR, SORT1, IL6R).

Polygenic scores of this theme: Abdominal aneurysm (29 variants) (PGS000753): the 26th percentile

Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
LRP1	rs1466535	GA	chip	G	1	0.70	LRP1: abdominal aortic aneurysm. (moderate evidence)	Typical / no judgement
DAB2IP	rs7025486	GG	chip	A	0	0.22	DAB2IP: aortic aneurysm and arterial disease. (moderate evidence)	Typical / no judgement
SORT1	rs602633	TG	chip	T	1	0.21	SORT1: aneurysm of the aorta (via LDL). (weak evidence)	Typical / no judgement
LINC00540	rs9316871	AG	chip	G	1	0.21	13q12: aortic aneurysm. (weak evidence)	Typical / no judgement
SMAD3	rs56062135	CC	chip	—	—	—	SMAD3 (TGF-β): aneurysm and dissection. (weak evidence)	Typical / no judgement
CDKN2B-AS1	rs10757274	GG	chip	G	2	0.50	9p21 is also the main common locus of abdominal aortic aneurysm. (moderate evidence)	Unfavourable

Favourable

• None.

Unfavourable

• CDKN2B-AS1 rs10757274 GG

Interpretation: Score at the 26th percentile (below the average); DAB2IP without risk alleles and LRP1 heterozygous (typical); IL6R Ala358 (protector) and LDLR T. The 9p21 homozygous is the weakness here too. The score does not include the dominant environmental factor (tobacco), absent in Sérgio F according to the profile.

Cardiac and myocardial insufficiency

Heart failure and cardiac muscle disease: sarcomere chaperones (BAG3, HSPB7) and transthyretin amyloidosis (TTR).

Polygenic scores of this theme: Heart failure (39 variants) (PGS003969): the 22nd percentile

Typical / no judgement

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
BAG3 C151R	rs2234962	TT	chip	C	0	0.22	Co-chaperone BAG3: common allele associated with dilated cardiomyopathy. (moderate evidence)	Typical / no judgement
CLCNKA/HSPB7	rs10927887	AG	imputation DR2=1.00	—	—	—	CLCNKA/HSPB7: heart failure. (weak evidence)	Typical / no judgement
HSPB7	rs1739843	TC	chip	—	—	—	HSPB7 (small heat shock protein): cardiomyopathy (weak evidence)	Typical / no judgement
TTR V122I	rs76992529	GG	chip	A	0	0.00	Transthyretin V122I: hereditary cardiac amyloidosis (rare in the reference population). (strong evidence)	Favourable
TTR V30M	rs28933979	GG	chip	A	0	—	Transthyretin V30M: hereditary amyloidosis (rare). (strong evidence)	Favourable
KLHL3/FHOD3	rs12460541	AA	imputation DR2=0.99	G	0	0.64	Heart failure (GWAS). (weak evidence)	Favourable

Favourable

- TTR rs76992529 GG
- TTR rs28933979 GG
- KLHL3/FHOD3 rs12460541 AA

Unfavourable

- None.

Interpretation: Score at the 22nd percentile (below the average). BAG3 without the allele of dilated cardiomyopathy. TTR V122I and V30M (hereditary myloidosis) absent. CLCNKA/HSPB7 heterozygous, without direction of clear effect. Truncant variants of titin (TTN), frequent cause of dilated cardiomyopathy, are not covered by the chip.

Stroke

Stroke (mainly ischemic). Atrial fibrillation, Lp(a) and coagulation also weigh here.

Polygenic scores of this theme: Stroke (90 variants) (PGS000038): the 19th percentile Favourable

Gene	SNP	Genotype (+)	Source	Effect allele	Copies	Freq.	Effect	Judgement
HDAC9	rs2107595	GG	chip	A	0	0.16	HDAC9: Stroke of large vessels. (moderate evidence)	Typical / no judgement
NINJ2	rs12425791	GA	chip	A	1	0.21	NINJ2: stroke.	Typical / no judgement
ZFHX3	rs879324	GG	chip	A	0	0.17	ZFHX3: Cardioembolic stroke. (weak evidence)	Typical / no judgement

Favourable

- None.

Unfavourable

- None.

Interpretation: Stroke score at the 19th percentile (favourable); HDAC9 (large vessels stroke) and ZFHX3 without risk; NINJ2 heterozygous (weak). With atrial fibrillation at the 2nd percentile and low genetic Lp(a), the genetic cardioembolic risk is low.

Mechanisms of ageing

Each finding was linked to the aging mechanisms of the list of objectives for immortality (84 mechanisms). The table shows only the mechanisms played by this report.

Mechanism	Unfavourable	Favourable	Interpretation
Vascular Aging	9p21 homozygous; APOA5, TCF21; thrombosis score in p86	metaGRS p5; Lp(a) without risk alleles; FA p2; Stroke p19; APOE ε3/ε3	Mixture: focal weakness in the arterial wall, favourable polygenic basis.
Cell Senescence	9p21 (CDKN2A/B = p16INK4a/p15INK4b; ANRIL) homozygous	—	9p21 regulates the senescence of vascular smooth muscle; it is the most direct link of this report with senescence.
Arterial stiffness	GUCY1A1 GG (via NO; common genotype)	PHACTR1 AA (protector)	Slightly favourable.
Chronic Inflammation / Inflammation	—	IL6R Asp358Ala (less IL-6); SH2B3 without risk	Slightly favourable.
Remodelling of the Extracellular Matrix / Fragmentation of Elastin	9p21 (aneurism)	P26 aneurysm score; DAB2IP without risk	Slightly favourable, except 9p21.
Metabolic Homeostasis Decline	APOA5 GC	LDLR GT; APOE ε3/ε3	Neutral (the lipid profile is in the other reports).
Intercellular Communication	F11 rs2289252 TT; thrombosis score p86	F5 and F2 negative; ABO probable O	Coagulation: moderate polygenic risk without major thrombophilia.
Protein Aggregation / Failure of Molecular Chaperones	—	TTR V122I/V30M absent; BAG3 without risk	Favourable in the heart.
Mitochondrial Calcium Deregulation	PLN TT (QT)	KCNN3 heterozygous, without clear effect	Small effect.
Clonal hematopoiesis (CHIP)	—	SH2B3 R262W absent	No visible germinal predisposition in this locus.
Deterioration of the Nuclear Blade and the Domains Associated with the Blade	—	SYNE2 heterozygous (typical)	No relevant effect.

Practical suggestions

All the suggestions below are suggestion (not applied): only from genetics, they are aligned with the v2 genetic plan and do not change the current stack without Sérgio F decision and the doctor.

Topic	Suggestion (not applied)	Genetic basis
9p21 / arterial wall	Keep LDL and ApoB as low as possible, without lower limit (the plan lipid-lowering strategy already does). More fiber, always (the better), and many raw vegetables and fruit every day. <i>suggestion (not applied)</i>	9p21 homozygous (independent effect of LDL, attenuated by diet rich in vegetables and fruit)
9p21 / senescence	The plan's monthly senolytic pulse (only fisetin + quercetin, and only with normal ALT) has an extra rationale here: 9p21 encodes p16INK4a, a senescence marker. <i>suggestion (not applied)</i>	CDKN2A/B (9p21)
Coronary image	Coronary calcium score (CAC) at the predicted age in the plan, to see if 9p21 is expressing itself in the arterial wall. <i>suggestion (not applied)</i>	9p21 homozygous
Venous thrombosis	In trips of more than 4 h: standing and walking every 1–2 h, drinking water, compression stockings. Always inform the doctor of this predisposition before surgeries, immobilizations or any hormone with estrogen or testosterone at a high dose. <i>suggestion (not applied)</i>	F11 TT, FGG, thrombosis score p86 (without Leiden or prothrombin)
Hydration	With dapagliflozin (diuresis) and retatrutide (less intake, possible gastrointestinal losses), ensure daily hydration, especially on hot days, long training or travel. <i>suggestion (not applied)</i>	Predisposition for venous thrombosis
Without natural anticoagulants	Do not add nattokinase, lumbrokinase or high doses of vitamin E to fish oil that already takes: add antiplatelet effect without proven benefit. <i>suggestion (not applied)</i>	Coagulation; CYP2C9 *1/*3 (profile)
Drugs that extend QT	If one day needs macrolides, fluoroquinolones, ondansetron, antipsychotics, methadone or antiarrhythmics, ask for ECG before and after (KCNE1 D85N was not seen in the data). Lithium of 5 mg is a very low dose. <i>suggestion (not applied)</i>	QT p36; KCNE1 unknown
Via NO-GMPc	Tadalafil acts downstream of guanylate cyclase (increases cGMP), which is consistent with the risk GUCY1A1. Hypothesis, without direct evidence of coronary benefit. <i>suggestion (not applied)</i>	GUCY1A1 rs7692387 GG
Atrial fibrillation	Nothing specific: very low genetic risk. Zero alcohol (already fulfilled) and regular sleep. <i>suggestion (not applied)</i>	FA p2 score
Omega-3 and vitamin D	Maintain omega-3 3.2–4 g/day (fish oil) and D3 10 000 IU as in the plan; target ferritin < 100 (iron excess is pro-oxidant for vessels). <i>suggestion (not applied)</i>	Coordination with the plan
What not to do	Do not use telomerase activators. Pausing curcumin, luteolin, apigenin and quercetin while the ALT is high is a suggestion (not applied; the real stack did not change) (except the senolytic pulse, only with normal ALT). <i>suggestion (not applied)</i>	V2 plan rules

Limitations

- The number of repetitions KIV-2 of LPA (main determinant of Lp(a)) is not measured by chips; SNPs are only markers.
- Rare large-effect variants are not covered: familial hypercholesterolemia (LDLR, APOB, rare PCSK9), protein C/S/antithrombin deficits, channelopathies (long/short QT syndromes, Brugada), truncation variants of titin and hypertrophic cardiomyopathies.
- Indels (PAI-1 4G/5G, ABO O allele) are not read reliably; group O is inferred by a tag (rs505922).
- The effect alleles of several SNPs come from GWAS Catalog (strand + in recent studies); the individual effects are small.
- The percentiles compare with the reference population (n=633); the predictive capacity of the scores is moderate (AUC typically 0.6–0.7).
- Only genes: the real risk depends heavily on lifestyle and drugs, which this report does not evaluate.

Methods and sources

- Genotypes: raw chip DNA file (GRCh38, strand +), analyzed only locally; Missing SNPs (Beagle 5.5, public reference panel of complete 30x genomes), accepted with $DR2 \geq 0.8$.
- Public positions and alleles: Ensembl REST (only rsIDs were consulted; no genotype left the computer).
- Effect alleles: cured from the literature and confirmed, when possible, in GWAS Catalog (local copy of 13-09-2026).
- Polygenic scores: PGS Catalog (harmonized GRCh38 files): PGS000018 (Inouye 2018), PGS003725 (Patel 2023), PGS000010 (Mega 2015), PGS000667 and PGS000752 (Lp(a)), PGS000043 (Klarin 2019), PGS002235 (Kolin 2021), PGS004920 (Wang 2021), PGS000768 (Lahrachi 2020), PGS000753 (Klarin 2020), PGS003969 (Rasoally 2023), PGS000038 (Rutten-Jacobs 2018). Percentages vs. reference population (n=633) with exactly the same variants.
- Main references: Clarke 2009 (LPA, NEJM); Klarin 2019 (vein thrombosis, Nat Genet); Lindström 2019 and Thibord 2022 (vein thrombosis); Aragam 2022 and van der Harst 2018 (coronal disease); From 2011 (9p21 × diet, PLoS Med); IL6R MR Consortium 2012 (Lancet); Cohen 2006 (PCSK9, NEJM); Roselli 2018 (FA, Nat Genet); Arking 2014 and Young 2022 (QT); Jones 2017 (aneurysm); Shah 2020 (heart failure); Malik 2018 (stroke).
- Mapping for 84 aging mechanisms of the target list. Report generated locally (HTML → PDF).